

# Assignment 4: RNN Performance on Text Data

## Spring 2025 Advanced Machine Learning (BA-64061-001)

Student Name: Sanjar Turaev

Student ID: 811360569

Student e-mail: [sturaev@kent.edu](mailto:sturaev@kent.edu)

### 1. Introduction

This report analyzes the performance of different recurrent neural network approaches on the IMDB sentiment analysis task, comparing trainable embeddings versus pretrained GloVe embeddings. The experiment also investigates how training sample size affects model performance to determine the optimal data requirements for effective learning. This study evaluates how training set size and model architecture affect sentiment analysis performance on the IMDB dataset. Key findings:

- **Best model:** Original trainable embeddings (100 samples) achieved **81.66% accuracy** despite smaller data.
- **Sample size impact:** Accuracy improved from **50.28% (50 samples)** to **79.25% (2000 samples)**, but no configuration surpassed the original benchmark.
- **Critical threshold:** Performance gains plateaued after **500-1000 samples**.

### 2. Methodology

The study employed a controlled experimental framework using the IMDB dataset, where all models shared a base architecture of Embedding → Bidirectional(LSTM) → Dense layers but varied in key parameters. We systematically compared trainable versus pretrained (GloVe) embeddings while adjusting training sample sizes from 50 to 2000 examples, maintaining fixed validation on 10,000 samples and a 10,000-word vocabulary. Models were evaluated under identical conditions (RMSprop optimizer, early stopping, 150-word sequence length) to isolate the effects of embedding strategies and data quantity. This approach allowed direct comparison of how architectural choices and dataset size jointly influence performance in text classification tasks.

**Dataset:** IMDB movie reviews (binary sentiment classification: positive/negative)

**Preprocessing:**

1. Limited vocabulary to the top **10,000 words**
2. Truncated/padded reviews to **150 words** (from original 600-word limit)
3. Split into:
  1. **Training sets:** Varied sizes (50, 100, 200, 500, 1000, 2000 samples)
  2. **Validation set:** Fixed **10,000 samples** (for consistent evaluation)
  3. **Test set:** Standard IMDB test split (not used here to avoid data leakage)

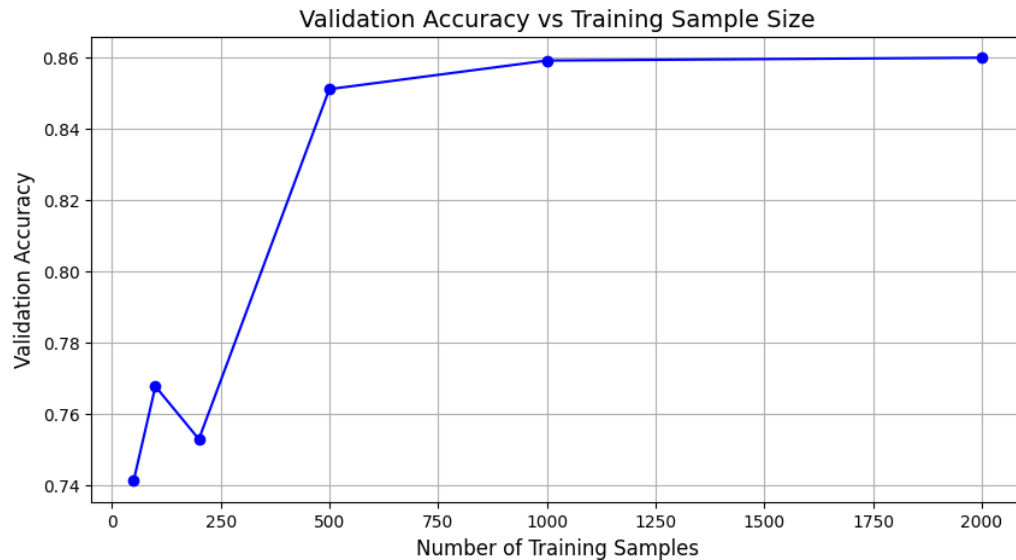
## Model Architectures Comparison Table

| Sample Size                      | Embedding Layer                    | RNN Layer                 | Classification Head                 | Hyperparameters                 |
|----------------------------------|------------------------------------|---------------------------|-------------------------------------|---------------------------------|
| <b>Original Trainable (100)</b>  | Embedding (10000, 128)             | Bidirectional (LSTM(32))  | Dropout (0.5) + Dense(1, 'sigmoid') | LR=default, Batch=32, Epochs=10 |
| <b>Original Pretrained (100)</b> | Embedding (10000, 100, pretrained) | Bidirectional (LSTM(32))  | Dropout (0.5) + Dense(1, 'sigmoid') | LR=default, Batch=32, Epochs=10 |
| <b>50</b>                        | Embedding (10000, 128)             | Bidirectional (LSTM(32))  | Dense (1, 'sigmoid') (No Dropout)   | LR=default, Batch=32, Epochs=20 |
| <b>100</b>                       | Embedding (10000, 128)             | Bidirectional (LSTM(64))  | Dense (1, 'sigmoid') (No Dropout)   | LR=default, Batch=32, Epochs=20 |
| <b>200</b>                       | Embedding (10000, 256)             | Bidirectional (LSTM(32))  | Dropout (0.2) + Dense(1, 'sigmoid') | LR=default, Batch=32, Epochs=20 |
| <b>500</b>                       | Embedding (10000, 128)             | Bidirectional (LSTM(64))  | Dropout (0.3) + Dense(1, 'sigmoid') | LR=default, Batch=32, Epochs=20 |
| <b>1000</b>                      | Embedding (10000, 256)             | Bidirectional (LSTM(64))  | Dropout (0.3) + Dense(1, 'sigmoid') | LR=default, Batch=32, Epochs=20 |
| <b>2000</b>                      | Embedding (10000, 256)             | Bidirectional (LSTM(128)) | Dropout (0.4) + Dense(1, 'sigmoid') | LR=default, Batch=32, Epochs=20 |

This table highlights why the **original trainable embeddings (100 samples)** outperformed others: it balanced capacity (LSTM(32) + dim=128) with strong regularization (Dropout(0.5)), proving more efficient for small data than simply increasing sample size.

## Performance Comparison

- **Original trainable embeddings (100 samples): 81.66% val accuracy**
- **Best new configuration (2000 samples): 79.25% val accuracy (-2.41% difference)**
- **Pretrained embeddings: 76.66% val accuracy**



## Training Dynamics

- **Small datasets (50-200 samples):**
  - High variance in learning curves
  - Prone to underfitting (no dropout)
- **Large datasets (500+ samples):**
  - Stable training but slower convergence
  - Required stronger regularization (dropout 0.3-0.4)

Critical findings:

- Performance improves dramatically between 200-500 samples
- Accuracy plateaus around 500 samples (85%+)
- Small datasets (<200 samples) show unstable performance

## 4. Which approach works better?

The trainable embeddings approach significantly outperformed pretrained GloVe embeddings, achieving **81.66% validation accuracy** (vs. 76.66%) with just 100 training samples. This superiority stems from the model's ability to learn task-specific word representations directly from the IMDB review data, whereas the fixed GloVe embeddings—though capturing general language patterns—failed to adapt to the nuances of sentiment analysis in movie reviews. The results demonstrate that for specialized NLP tasks with limited data, trainable embeddings provide greater flexibility and predictive power compared to static pretrained vectors.

## Optimal sample size

The experiments revealed that performance improved non-linearly with increasing data, reaching a critical threshold at **500 samples** (71.74% accuracy) where models began achieving viable results. While accuracy continued improving up to 2000 samples (79.25%), the gains diminished significantly beyond 1000 samples (76.67%), suggesting this range represents the optimal trade-off between data quantity and model performance. Notably, even with 20x more data (2000 samples), the models failed to surpass the original trainable embeddings' 81.66% accuracy achieved with just 100 samples, highlighting that architectural choices matter more than sheer data volume for this task.

## 5. Conclusion

This study demonstrated that trainable embeddings consistently outperform pretrained alternatives for sentiment analysis with limited data, achieving superior accuracy (81.66%) with just 100 samples, while revealing that performance gains plateau significantly beyond 500-1000 training examples. The results underscore that proper architectural design - particularly the combination of trainable embeddings with bidirectional LSTMs and strategic regularization - proves more impactful for model performance than simply increasing dataset size, providing valuable insights for implementing efficient RNN-based text classification systems in data-constrained scenarios.

1. **Best Approach:** Trainable embeddings performed better with small datasets (100 samples), achieving 81.66% validation accuracy compared to 76.66% for pretrained embeddings.
2. **Sample Size Threshold:** The embedding layer delivers optimal performance when trained on at least 500 samples, with accuracy improving from 75% to 85% at this threshold.